CPE 009B | OBJECT ORIENTED PROGRAMING 2

LABORATORY EQUIPMENT BORROWING AND INVENTORY MANAGEMENT SYSTEM

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WHAT THE APPLICATION DOES

A PyQt6 desktop application that replaces the paper borrowing slip with a digital workflow, taking each request from submission through professor approval, release and return, with all records saved in CSV files.



STUDENT

Searches the stock, fills the slip, submits the request and tracks its status



PROFESSOR

Reviews each request, approves or declines it with a reason, posts announcements



LABORATORY STAFF

Releases approved items with a Lab Ticket ID, confirms returns, manages stock



ITSO

Keeps the CSV log of every slip and creates or deletes backup folders

THE PROBLEM:
BACKGROUND

In borrowing laboratory equipment in a school, the current process starts when a student gets a borrowing slip from the laboratory staff. The student fills out the required information and asks the professor to sign the slip for approval. Once the professor has signed the slip, the student will give the approved slip and their ID to the laboratory staff. The staff checks the approval of the professor based on the signature, then release the requested materials, and records its release and return.

This method is inefficient. It is time-consuming for the student to fill out the form and wait for the professor's approval. The student may also find out only at the laboratory that the needed item is already borrowed or unavailable. Filling out the slip by hand can also lead to erasures, missing quantities, missing signatures, and incomplete information and also ITSO does not have any organized records about the slip. This project addresses these problems by replacing the paper slip with a desktop application, which makes the process more organized, optimal and keeps all records saved in one place.

MAIN PROBLEM

The manual borrowing slip system slows down the requesting and approval procedures for laboratory equipment, making it difficult for the institution to keep track of equipment availability, maintain accurate records of active borrowers, and ensure equipment is properly released and returned (Bishwas, 2014; RSIS International, 2026).

SPECIFIC PROBLEM

1. Requesting and approving a borrower slip takes significant time because students must manually complete paper forms, locate assigned professors for physical signatures, and line up at the laboratory stockroom counter (Bishwas, 2014; RSIS International, 2026).
2. Handwritten slips lead to erasures, missing material details, and incorrect quantities, causing students to discover that needed items are out of stock only upon arriving at the equipment counter (Bishwas, 2014; RSIS International, 2026).
3. There is no systematic record of each request, making it difficult for laboratory staff and administrators to trace who has borrowed which equipment or determine the real-time status of borrowed materials (Bishwas, 2014; RSIS International, 2026).

THE PROBLEM:
WHO IS AFFECTED AND WHY ADDRESS IT

Student

- Affected: Fills in the slip by hand and waits for a signature. May learn only at the laboratory that the item is gone.
- Address: Saves time and avoids wasted trips. Students can search the stock and track their request status online.

Professors

- Affected: Must review and sign every paper slip, so approval waits until the professor can be found.
- Address: Requests can be approved or declined anytime, with a reason, in one dashboard.

Laboratory Staff

- Affected: Checks signatures by eye and records every release and return by hand, with no clear view of what is available.
- Address: Prevents wrong releases. The app checks the stock and the approval, then updates the inventory automatically.

ITSO

- Affected: Has no organized records, so nobody can easily see who borrowed what.
- Address: Gives one reliable record. Every slip is saved in the CSV log, with backups for safety.

GENERAL OBJECTIVES

To develop a PyQt6 desktop application that will take a laboratory borrowing slip through a digital workflow which the student submit a student slip request, the request is saved at the ITSO Csv log, the request is approved or denied by the professor, and the laboratory staff release and retrieve the materials and maintain an accurate inventory.

SPECIFIC OBJECTIVE

1.To create role-based logins and different dashboards for students, Itso, professors and laboratory staff, and have a borrow-slip workflow (request, saved at csv log, approve or reject, update the inventory csv based what material borrow, release, return).

2.To validate all user entry (login credentials, student details about ID and name, quantities, required fields, equipment availability), and provide feedback to the student user via message and dialog boxes.

3.To store and load all of the students borrow slips requested in a CSV file to retain data records of all the requests. Additionally, having a CSV file for inventory so the data will retain even the application was closed

4.To apply the object-oriented principles of encapsulation, abstraction, inheritance, and polymorphism in creating the classes which included the user class, equipment class, slip, inventory management, data handling and also the window classes.

5.To package the application as a Windows executable (.exe) that will run without a Python installed.

SCOPE AND DELIMITATIONS

SCOPE

- Login System: role, username, password, show password; wrong entry shows an error and shakes the window
- Student Dashboard: fill in the slip, search stock, track requests, return to staff
- Professor Dashboard: approve or decline (reason required), post announcements
- ITSO Dashboard: CSV slip log, refresh, generate and delete backup folder
- Laboratory Staff Dashboard: add stock, release with Lab Ticket ID, confirm return

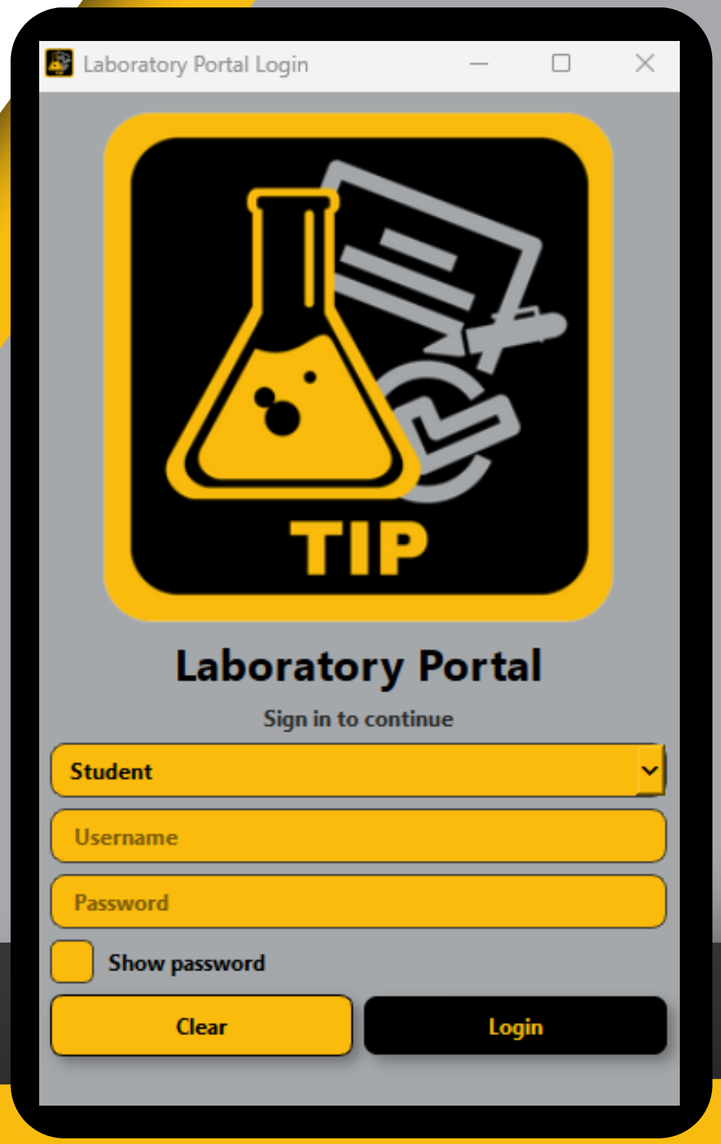
DELIMITATIONS

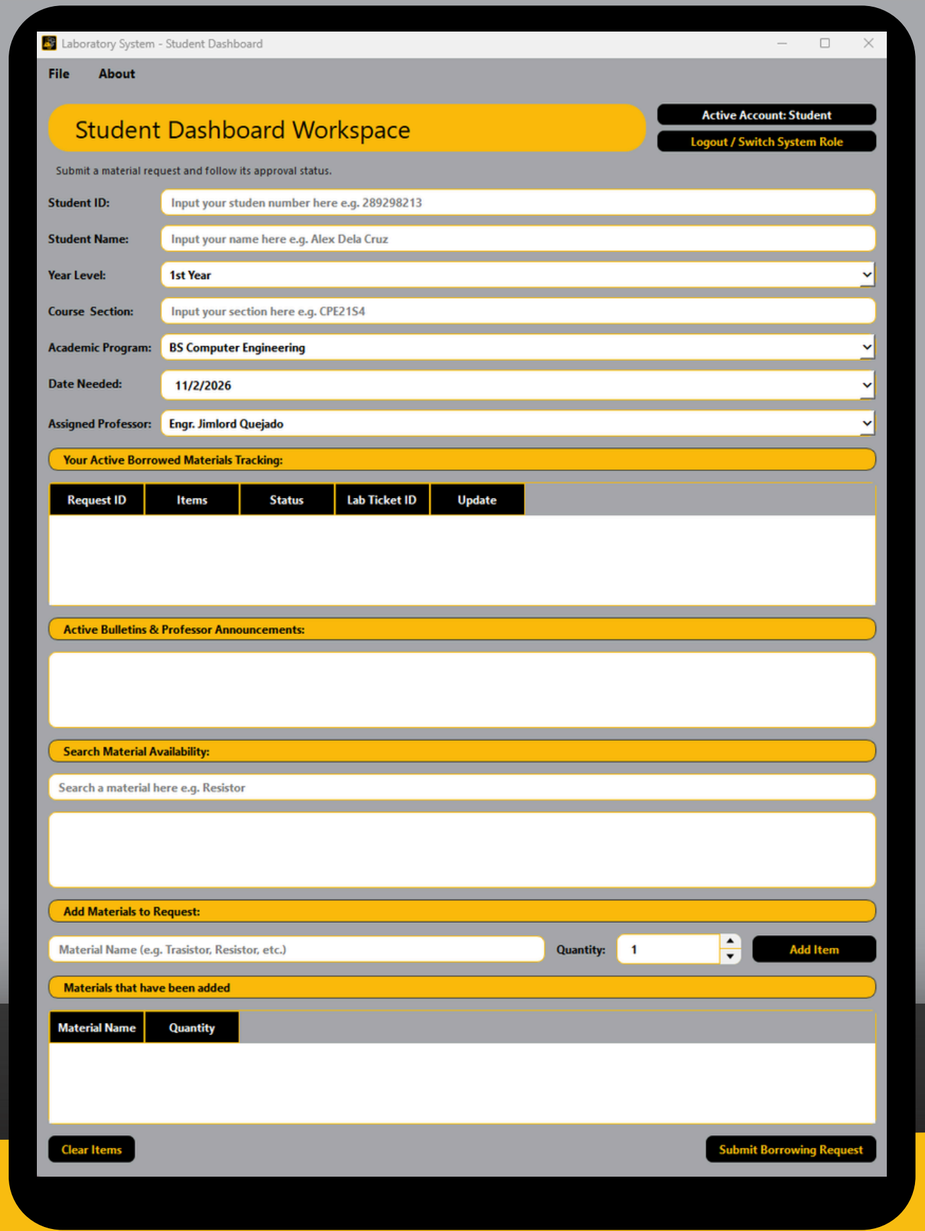
- No database (MySQL, MySQLite3, SQLite)
- Limited to TIP Students, Professors and Laboratory Staff
- Four demo accounts only; no full authentication or registration
- Staff still check and hold the physical ID
- Runs on a single computer; no syncing across computers
- No email or SMS; announcements appear in the app only
- Staff still inspect damaged, missing or incomplete materials
- No barcode or RFID
- No simultaneous requests from multiple devices
- A prototype, not a full multi-laboratory system

OBJECTIVES AND MATCHING FEATURES

#	Specific objective	Matching feature in the application
1	Role-based logins, four dashboards and the borrow-slip workflow	Login window; Student, Professor, Staff and ITSO dashboards; status path from request to return
2	Validate entries and give feedback through message and dialog boxes	Warning messages, Lab Ticket ID dialog, stock check before an item is released
3	Store and load slips and stock so data is kept after closing	ITSO_Maintenance_data.csv and Inventory_data.csv
4	Apply encapsulation, abstraction, inheritance and polymorphism	InventoryManager, FileReaderWriter and CSVFileReaderWriter, 17 classes in 15 modules
5	Package the application as a Windows .exe	Packaged with PyInstaller into a working Windows .exe that runs without Python installed

LOGIN AND STUDENT DASHBOARD





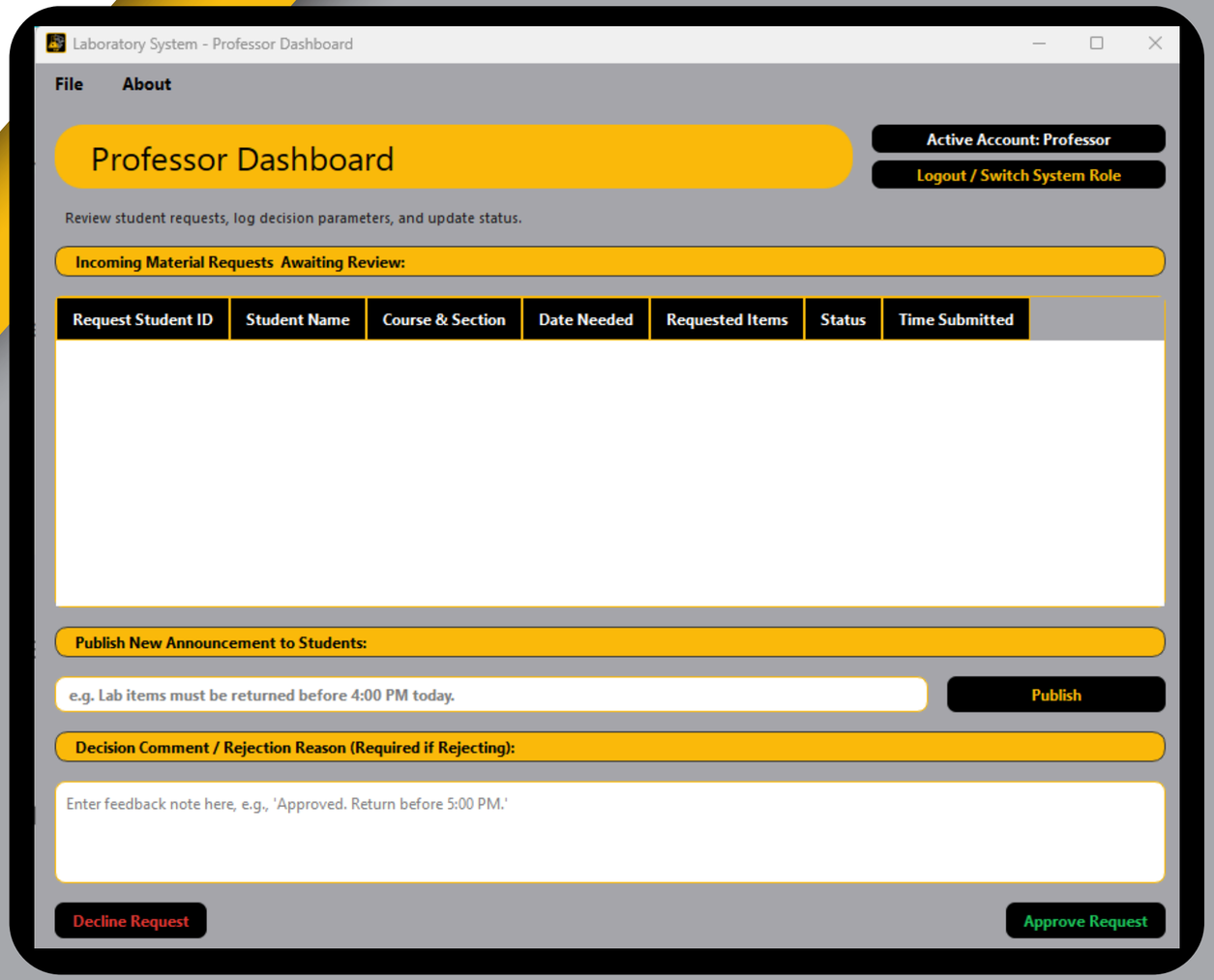
LOGIN WINDOW

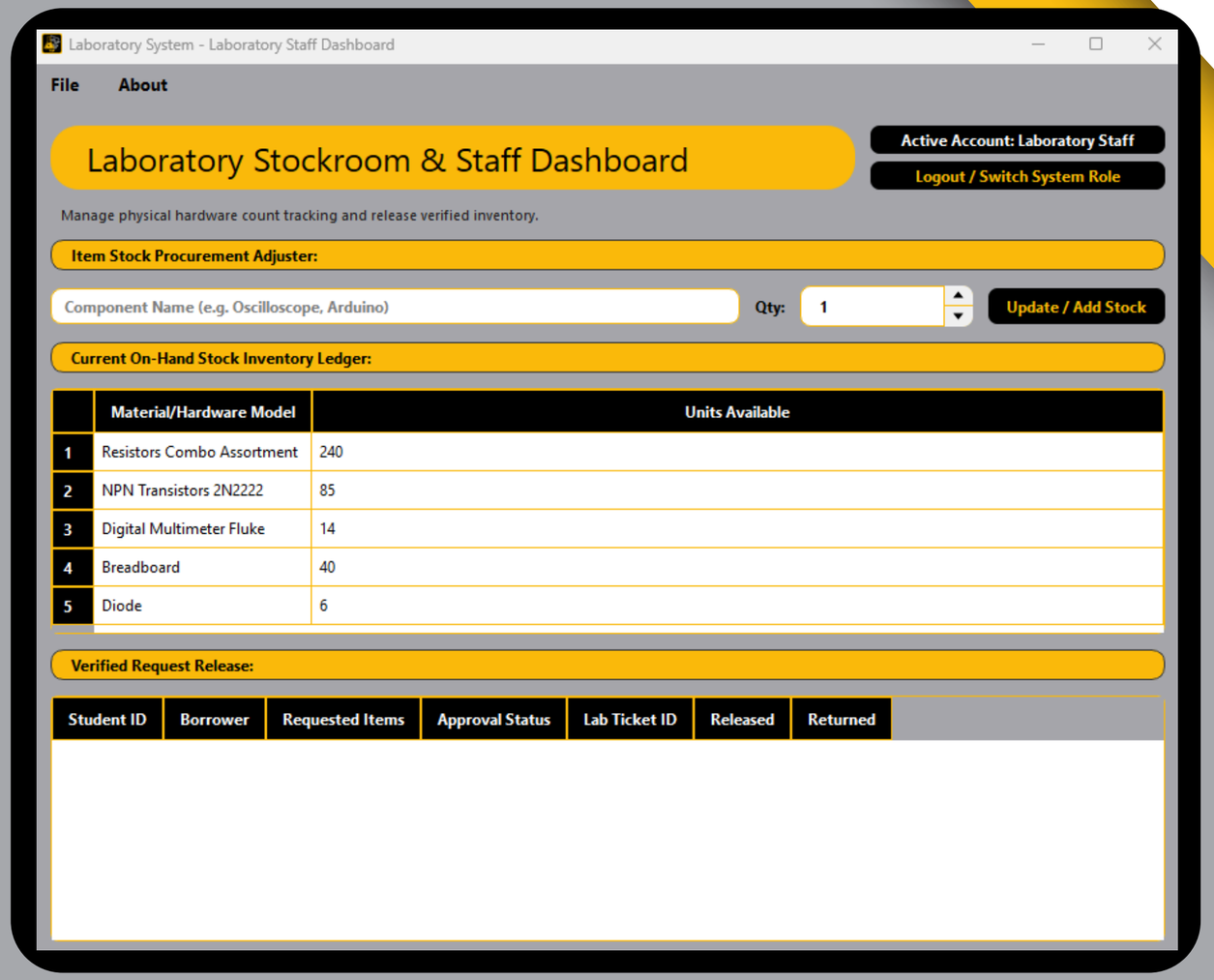
- Pick a role, enter username and password
- Wrong entry: red fields, error message, window shakes

STUDENT DASHBOARD

- Fill in the slip and add materials
- Search the stock for units available
- Follow the status and Lab Ticket ID
- Return to Staff when dones

PROFESSOR AND STAFF DASHBOARDS

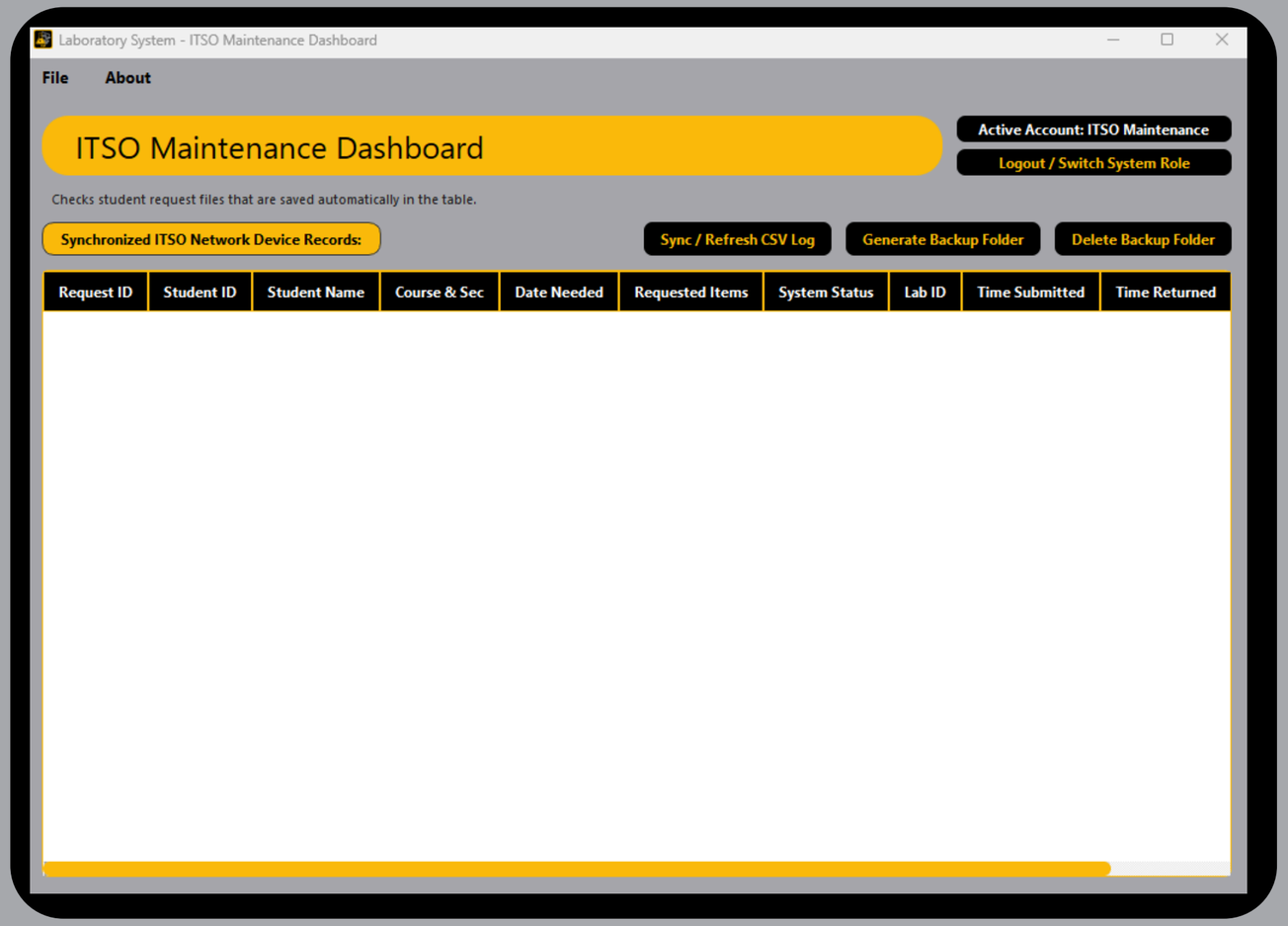




Professor: approve, decline (reason required), publish announcements

Staff: add stock, release with Lab Ticket ID, confirm return

ITSO DASHBOARD



Sync / Refresh CSV Log: Reloads every slip from the CSV file.

Generate Backup Folder: Copies the log into an ITSO Backup Folder.

Delete Backup Folder: Removes a backup folder selected by the user.

10-column slip log: Tracks details from Request ID to time returned, including system status and Lab ID.

OOP PRINCIPLE APPLIED

ABSTRACTION

Where applied: CSVFileReaderWriter class, read() and write() methods

When the dashboard wants to load data, it calls the csv_handler.read(filepath). The dashboard does not need to know how python opens a file, handles delimiters, or parses strings. The dashboard does not need to know the exact steps of how python opens a file, uses delimiters, or processes lists. It hides those unorganized details inside this method.

INHERITANCE

Where applied: CSVFileReaderWriter class (Child) inherits FileReaderWriter class (Parent)

When the dashboard wants to load data, it calls the csv_handler.read(filepath). The dashboard does not need to know how python opens a file, handles delimiters, or parses strings. The dashboard does not need to know the exact steps of how python opens a file, uses delimiters, or processes lists. It hides those unorganized details inside this method.

POLYMORPHISM

Where applied: CSVFileReaderWriter.read() overrides FileReaderWriter.read()

It allows a method to act differently depending on the specific object that calls it. The parent class FileReaderWriter has a default read() method that just prints a message and returns an empty list. The child class CSVFileReaderWriter overrides this method by using the exact same name but replacing the inside logic with instructions to analyze spreadsheet files. When the code calls read(), python runs the specific child version instead of the default parent layout.

OOP PRINCIPLE SOURCE CODE

INHERITANCE

```python
from FileReader import FileReaderWriter
import csv

    # this  file and function  is used for converting  the  borrowing  slip  of  the  student
    # from  the  student  dashboard  into csv  file and  received  to the itso dashboard

    # opens  a csv file  and  returns  all  lines as a list
class CSVFileReaderWriter(FileReaderWriter):
    def read(self, filepath):
        with open(filepath, newline='') as  csvfile:
            data = csv.reader(csvfile, delimiter=',', quotechar='|')
            return list(data)

    # opens  a csv  file and  adds a  new  line of  data to  the  end of  it
    def write(self, filepath, data):
    # opens the  file to  add new  student  lines without  deleting  the  old  ones
        with open(filepath, 'a', newline  = '') as csvfile:
            writer = csv.writer(csvfile,  delimiter=',', quotechar='|', quoting=csv.QUOTE_MINIMAL)
            writer.writerow(data)
```

ABSTRACTION

```python
class CSVFileReaderWriter(FileReaderWriter):
    def read(self, filepath):
        with open(filepath, newline='') as  csvfile:
            data = csv.reader(csvfile, delimiter=',', quotechar='|')
            return list(data)
```

POLYMORPHISM

```python
# create  our parent class
class FileReaderWriter():
    # base  layout for  reading  files  that returns an empty  list  if  not  changed
    def read(self):
        print("This is the default read method")
    # returns an  empty  list as a safety  backup to prevent the  program from crashing if no  data is  found
        return[]
    # base  layout for  writing data into files
    def write(self):
        print("This is the  default write method")
```

DEMONSTRATION OF WORKING APPLICATION

SELECTED TEST CASES

#	Feature	Action	Expected	Actual	Result
3	Login	Role does not match the account	Login rejected	Error message shown	Pass
7	Student slip	Submit with no items added	Warning	Validation error, nothing saved	Pass
14	Professor	Decline without a reason	Warning	Warning shown, status unchanged	Pass
23	Staff release	Release more than the available stock	Release blocked	Warning shown, nothing released	Pass
24	Staff release	Cancel the Lab Ticket ID dialog	Release cancelled	Warning shown, nothing released	Pass
25	Staff return	Press Confirm Return	Stock restored	Status cleared, units added back	Pass
30	File saving	Close and reopen the program	Data is still there	Slips and stock stored in CSV files	Pass

OBJECTIVES EVALUATION

#	Specific Objective	Met?	Evidence
1	Role-based logins, dashboards and the full borrow-slip workflow	Yes	Figures 1 to 11, Tests 1 to 5, 11, 16, 22, 25
2	Validate user entries and give feedback	Yes	Tests 1, 3, 6, 7, 8, 14, 17, 20, 23
3	Store and load slips and inventory in CSV files	Yes	Tests 11, 15, 16, 19, 21, 27, 30
4	Apply abstraction, inheritance and polymorphism	Partly	Sections 4.3 to 4.5
5	Package as a Windows .exe	Yes	Figure 12

CONCLUSION

The group built a PyQt6 desktop application called the Laboratory Equipment Borrowing and Inventory Management System. It replaces the paper borrowing slip with a digital process. A student submits a request, the professor approves or declines it, the laboratory staff release and receive the items, and ITSO keeps a record of every slip in a CSV file. The program also keeps the inventory updated when items are added, released and returned. It has a login for each role, a menu bar, dialogs, a QSS theme, and it was packaged as a Windows .exe. All 30 test cases passed, and four of the five objectives were fully met. Objective 2 was partly met because the student ID and name are only checked if they are empty.

The application answers the problem in the paper. Students no longer need to fill out a slip by hand or line up just to find out an item is not available, because they can search the stock and track the request. Professors can approve from the dashboard, and the staff can see the approved requests. ITSO has an organized record with the submitted and returned times, so it is easier to know who borrowed what. The OOP principles were also applied through the file reader classes, the data classes, the inventory manager, and the separate UI and logic classes.

RECOMMENDATIONS

Add email or SMS notifications. Students would know right away when a request is approved, declined or ready for release, so they don't need to keep checking the app.

Add a toolbar button for common actions, and a log for damaged or missing items. This would help the staff record problems when the items are returned.

Let each professor see only their assigned requests. Right now a professor sees every request, so this would remove confusion and protect the privacy of students.

Add stronger checks for the student ID, name and section. Checking the format would stop wrong or fake data from going into the ITSO records.

Add a database such as SQLite. CSV files are easy to open and change by mistake, and they can be damaged when two programs use them. A database is safer and faster when there are many records.

Add real accounts and registration. The usernames and passwords are fixed in the code. Real accounts would show which student or professor is using the app, and passwords would be safer.

Make the program work on many computers. The app runs on one computer only. A network or server version would let students request from their own devices and the staff release at the lab at the same time.

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